

Two reflections and exact ranks for level-four double polylogarithms

A proof of the manuscript's odd-weight rank conjecture,
an explicit parity compiler, and uniform harmonic-sum obstructions

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Abstract

The ProveIt manuscript records an all-odd-weight conjecture for a particular rational matrix of level-four depth-two shuffle and stuffle product relations. We prove both asserted ranks, not only their difference. If A_w is the matrix on its $6w - 7$ convergent imaginary coordinates, then $\text{rank } A_w = 5w - 6$ for odd $w \geq 3$, and $\text{rank } A_w = 6w - 7$ for even $w \geq 2$. After deleting the Gaussian columns, the corresponding ranks are $(9w - 11)/2$ and $5w - 6$. The proof converts the binomial rows into equations for six homogeneous polynomials. Two independent polynomial reflections then give an explicit integer basis of the entire odd-weight nullspace. This yields a complete row-membership test and proves that the Gaussian-only linear consequences are exactly the same-point shuffle consequences.

Keeping the single-value right sides gives a closed polynomial compiler for all even-weight imaginary double values at fourth roots of unity. We extract two compact mixed-color identities in every even weight. A second consequence is a uniform obstruction: in every odd weight, the sum $S_{2m} = \sum_{n \geq 0} (-1)^n H_n / (2n + 1)^{2m}$ introduces one additional formal direction beyond the Gaussian doubles in this restricted linear system. The proposed numerical evaluation of S_4 is neither proved nor refuted. Exact certificates cover weights 2–41, and 210 explicit even-weight formulas through weight 12 are checked against all 816 product rows. All ranks are formal; no numerical period independence is asserted.

Keywords. Multiple polylogarithms; colored double zeta values; double shuffle; Gaussian periods; exact linear algebra; parity.

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1 Research target, result, and scope

The source for this continuation is the consolidated ProveIt manuscript at commit 9bc738d3be22b8586a24693f19fb2e2a50ecd1bf. The target is the conjecture labeled `signed:conj:rank` in `chapters/05-signed-kernels.tex`; the underlying file has Git blob e61ff32263722d7575ca0202fba4dd7dad534861. That conjecture gives two ranks for an explicitly specified family of rational matrices, with exact experimental evidence through odd weight 31 [1].

This target is distinct from the manuscript's previously proved Gaussian weight-six formulas, its earlier shuffle-rank theorem, and its distribution rank results. The result below settles the remaining matrix conjecture and explains its experimental pattern. The method also covers the even weights and produces an explicit solution operator for the inhomogeneous equations.

Theorem 1.1 (All-weight rank theorem). *Let A_w be the level-four imaginary depth-two product matrix defined in Section 2. Let $A_w^{\neg g}$ be obtained by deleting all columns $\text{Im Li}_{a,w-a}(i, 1)$, $1 \leq a < w$. Then*

$$\text{rank } A_w = \begin{cases} 5w - 6, & w \geq 3 \text{ odd}, \\ 6w - 7, & w \geq 2 \text{ even}, \end{cases} \quad (1.1)$$

$$\text{rank } A_w^{\neg g} = \begin{cases} (9w - 11)/2, & w \geq 3 \text{ odd}, \\ 5w - 6, & w \geq 2 \text{ even}. \end{cases} \quad (1.2)$$

For odd w , an explicit integer basis of $\ker A_w$ has $w - 1$ members, split into two families of $(w - 1)/2$ members. One family controls the Gaussian coordinates; the other vanishes on every Gaussian coordinate.

The rank refers to coefficient rows after the known single-value right sides have been suppressed. It is not a dimension of numerical periods. If the evaluation map has further identities, those identities can reduce the numerical span without changing A_w . Similarly, an obstruction in this matrix system excludes only a rational linear combination of its stated rows. It does not exclude argument transformations, elimination from higher depth, or products of lower-weight relations followed by further reduction.

The even-weight reduction phenomenon belongs to the established parity theory of multiple polylogarithms, notably Panzer's theorem [3]. Our contribution here is an explicit description of this particular linear product module and its compiler; we do not claim a new general parity principle. Standard cyclotomic relation systems are substantially broader than this module [2]. The literature search also located recent work on a different, weight-two conjecture of Zhao [4]; it should not be conflated with the manuscript conjecture proved here. No exhaustive claim of worldwide priority is made for individual special-value identities.

2 The exact product model

2.1 Colors, indices, and convergent coordinates

Our convention is

$$D_{a,b}(x, y) = \text{Li}_{a,b}(x, y) = \sum_{n > m \geq 1} \frac{x^n y^m}{n^a m^b}, \quad a, b \geq 1. \quad (2.1)$$

Thus a is the outer index. We use

$$H_n^{(b)} = \sum_{j=1}^n j^{-b}, \quad H_n = H_n^{(1)}, \quad H_0^{(b)} = 0, \quad \beta(k) = \sum_{j \geq 0} \frac{(-1)^j}{(2j+1)^k} \quad (k \geq 1),$$

and $\zeta(k) = \sum_{n \geq 1} n^{-k}$ for $k \geq 2$. The series for $\beta(1)$ is understood in its ordinary alternating sense. Put

$$x_{a,b}^{r,s} = \text{Im } D_{a,b}(\mathbf{i}^r, \mathbf{i}^s), \quad r, s \in \mathbb{Z}/4\mathbb{Z}.$$

Conjugation gives $x_{a,b}^{r,s} = -x_{a,b}^{-r,-s}$. When both colors are even, the imaginary coordinate is zero. The six remaining color representatives are

$$(0,1), (1,0), (1,1), (1,2), (1,3), (2,1). \quad (2.2)$$

For each representative, retain $a = 1, \dots, w-1$, except for $x_{1,w-1}^{0,1}$. This last coordinate has a logarithmically divergent imaginary part. There are consequently $6(w-1) - 1 = 6w - 7$ retained coordinates. We order them lexicographically by (a, b, r, s) , with $b = w - a$.

For $a > 1$, absolute convergence follows from $H_{n-1} = O(\log n)$ and $H_{n-1}^{(b)} \leq H_{n-1}$. For $a = 1$ and outer color different from 1, summation by parts gives the usual convergent boundary values. For example, when the inner color is 1, $H_{n-1}^{(b)}/n$ tends to zero and has summable eventual variation; when the inner color is nontrivial, its partial sums converge to a finite single value with an $O(n^{-b})$ remainder. The exceptional outer color 1 and imaginary inner color produces the removed logarithm. Purely real divergent expressions are not assigned finite numerical values: their imaginary equations are simply zero.

2.2 The row vocabulary

The shuffle product is

$$D_{p,q}(x, y) + D_{q,p}(y, x) = \text{Li}_p(x) \text{Li}_q(y) - \text{Li}_w(xy), \quad p + q = w. \quad (2.3)$$

The indices *and the colors* exchange together. The product shuffle is

$$\begin{aligned} & \sum_{j=0}^{p-1} \binom{q+j-1}{j} D_{q+j, p-j}(y, x/y) \\ & + \sum_{j=0}^{q-1} \binom{p+j-1}{j} D_{p+j, q-j}(x, y/x) = \text{Li}_p(x) \text{Li}_q(y). \end{aligned} \quad (2.4)$$

The first identity follows by partitioning the product of two sums into $n > m$, $m > n$, and $m = n$. The second is the iterated-integral shuffle, with the binomial coefficients counting interleavings of the zero letters; its coefficient form is also checked directly in Section 3. The boundary and regularization conventions agree with the source model and standard regularized double-shuffle relations [1, 2].

For each $p + q = w$ and $r, s \in \mathbb{Z}/4\mathbb{Z}$, take imaginary parts with $x = \mathbf{i}^r$, $y = \mathbf{i}^s$. If neither factor is $\text{Li}_1(1)$, retain both coefficient rows. If exactly one factor is $\text{Li}_1(1)$, retain only the *shuffle-minus-shuffle* row. Its divergent double coordinate cancels and its right side is $-\text{Im } \text{Li}_w(xy)$. If both factors are divergent, the imaginary equation is zero. Discard zero rows; retain duplicates if desired. This is A_w . Repeated rows affect neither rank nor row membership.

For example, the implementation with repetitions has 92 rows and 23 columns at weight 5, exactly matching the canonical manuscript model. We do not add distribution relations, arbitrary changes of argument, or higher-depth relations. Suppressing the single-value right sides gives the homogeneous coefficient system used in the rank proof.

2.3 A formal quotient and its evaluation meaning

Let $\mathcal{V}_w = \mathbb{Q}^{6w-7}$, viewed as row vectors of formal double-coordinate coefficients, and set

$$\mathcal{R}_w = \text{Row}(A_w), \quad \mathcal{Q}_w = \mathcal{V}_w / \mathcal{R}_w.$$

The quotient suppresses the single-value right sides of the specified rows. It is deliberately not the algebra of all colored periods modulo all products. A vector in $\mathcal{R}_w$ yields a valid reduction to a particular linear combination of the displayed single-value right sides. The converse need not hold for true numerical identities. This separation is essential when using $\mathcal{Q}_w$ to reject an insufficient proof search.

3 Completing the missing coordinate and encoding the rows

Set $N = w - 2$, and introduce six homogeneous polynomials of degree N :

color	(0,1)	(1,0)	(1,1)	(1,2)	(1,3)	(2,1)
polynomial	A	B	C	D	E	F.

(3.1)

For example, $B(X, Y) = \sum_{a+b=w} v_{a,b}^{1,0} X^{a-1} Y^{b-1}$ for a candidate null vector v . The coefficient absent from the matrix occurs in A.

Lemma 3.1 (Unique homogeneous completion). *Every vector in $\ker A_w$ extends uniquely to six complete polynomials satisfying all homogeneous shuffle and stuffle equations. The missing coefficient is assigned*

$$v_{1,w-1}^{0,1} = -v_{w-1,1}^{1,0}. \quad (3.2)$$

Restriction to the retained coordinates is an isomorphism back to $\ker A_w$.

Proof. Assigning (3.2) makes the missing stuffle equation zero. Its shuffle equation is then zero because the retained difference equation is zero. This handles the two orders of the divergent factor and their conjugates. The absent coefficient occurs nowhere else: in a shuffle sum it can have outer index 1 only when $q = 1, j = 0$ or $p = 1, j = 0$, precisely the divergent-factor cases. All other equations were already retained. Conversely, a complete homogeneous solution satisfies every retained row. The missing stuffle equation forces (3.2), giving uniqueness. The argument includes $w = 2$; the equation with two divergent factors is purely real. $\square$

For a homogeneous polynomial P , write $P^\top(X, Y) = P(Y, X)$. The stuffle equations reduce to

$$A = -B^\top, \quad C = -C^\top, \quad D = -F^\top, \quad E = E^\top. \quad (3.3)$$

For instance, swapping the colors (1,3) gives (3,1), which is minus (1,3) under conjugation; this explains the positive symmetry of E .

Lemma 3.2 (Polynomial form of the shuffle rows). *The remaining equations are exactly*

$$E(X + Y, X) + A(X + Y, Y) = 0, \quad (3.4)$$

$$B(X + Y, X) + B(X + Y, Y) = 0, \quad (3.5)$$

$$C(X + Y, Y) - F(X + Y, X) = 0, \quad (3.6)$$

$$D(X + Y, Y) - D(X + Y, X) = 0. \quad (3.7)$$

Together with (3.3), these equations include all colors and all weight splits, including the completed boundary equations.

Proof. For general color polynomials $P_{r,s}$, the polynomial whose coefficient of $X^{p-1}Y^{q-1}$ is the shuffle row is

$$P_{s,r-s}(X+Y, X) + P_{r,s-r}(X+Y, Y).$$

Indeed, the term with indices $(a, b) = (q+j, p-j)$ in the first summand has the coefficient $\binom{q+j-1}{q-1} = \binom{q+j-1}{j}$, exactly as in (2.4). The second summand gives the other binomial sum. Use the color pairs $(0, 1)$, $(1, 1)$, $(1, 2)$, and $(1, 3)$ in turn. Their polynomials are the four displayed left sides. The pairs $(1, 0)$ and $(2, 1)$ just exchange X, Y in equations already obtained. Conjugate pairs multiply an equation by -1 ; even-even pairs are zero. This exhausts the possibilities. $\square$

The binomial matrices have now disappeared. What remains is a pair of reflection problems for ordinary polynomials.

4 The two reflection blocks

4.1 The block carrying Gaussian coordinates

From (3.4) and $A = -B^\top$, the invertible change of variables $(X, Y) \mapsto (X+Y, X)$ gives

$$E(X, Y) = B(X-Y, X). \quad (4.1)$$

Equation (3.5) becomes

$$B(X, Y) + B(X, X-Y) = 0. \quad (4.2)$$

Let $\varepsilon = (-1)^N$. Homogeneity and (4.2) imply

$$\begin{aligned} E(Y, X) &= B(Y-X, Y) = \varepsilon B(X-Y, -Y) \\ &= -\varepsilon B(X-Y, X) = -\varepsilon E(X, Y). \end{aligned}$$

Because $E = E^\top$, it follows that $(1+\varepsilon)E = 0$. If N is even, $E = 0$, hence $B = 0$ and $A = 0$. If N is odd, the symmetry of E imposes no additional condition.

When $N = 2m - 1$ is odd, the change of variables $U = X$, $V = 2Y - X$ turns (4.2) into oddness in V . Consequently the complete Gaussian block has the basis

$$B_j(X, Y) = X^{N-2j-1}(2Y-X)^{2j+1}, \quad 0 \leq j < m, \quad (4.3)$$

with $A_j = -B_j^\top$ and

$$E_j(X, Y) = (X-Y)^{N-2j-1}(X+Y)^{2j+1}. \quad (4.4)$$

Its dimension is exactly m .

4.2 The complementary block

Equations (3.6) and (3.3) give

$$F(X, Y) = C(X, X-Y), \quad D(X, Y) = -C(Y, Y-X), \quad (4.5)$$

where $C = -C^\top$. The last shuffle equation asks for $D(X, Y) = D(X, X-Y)$. But

$$\begin{aligned} D(X, X-Y) &= -C(X-Y, -Y) \\ &= -\varepsilon C(Y-X, Y) = \varepsilon C(Y, Y-X) = -\varepsilon D(X, Y). \end{aligned}$$

Thus $(1 + \varepsilon)D = 0$. If N is even, $D = 0$ forces $C = F = 0$. If N is odd, the only condition is $C = -C^\top$. A basis is

$$C_j(X, Y) = X^j Y^{N-j} - X^{N-j} Y^j, \quad 0 \leq j < m. \quad (4.6)$$

This block also has dimension m and vanishes identically on the Gaussian polynomial B .

Theorem 4.1 (Complete nullspace normal form). *In every even weight $w \geq 2$, $\ker A_w = 0$. In odd weight $w = 2m + 1$, set $N = 2m - 1$. Every vector in $\ker A_w$ is obtained uniquely by choosing*

$$\begin{aligned} B &\in \text{span}_{\mathbb{Q}}\{B_0, \dots, B_{m-1}\}, \\ C &\in \text{span}_{\mathbb{Q}}\{C_0, \dots, C_{m-1}\}, \end{aligned}$$

then setting

$$(A, B, C, D, E, F) = (-B^\top, B, C, -C(Y, Y - X), B(X - Y, X), C(X, X - Y)), \quad (4.7)$$

and dropping the missing coefficient of A . All resulting basis vectors have integer coordinates.

Proof. The preceding two block computations prove necessity and show that the two blocks have disjoint determining polynomials. Substitution into (3.3)–(3.7) proves sufficiency. The bases (4.3) and (4.6) are independent respectively by the change to (U, V) and by their disjoint antisymmetric monomial pairs. Deleting the missing coefficient cannot destroy independence, because the B and C polynomials are retained in full. Lemma 3.1 finishes the argument. $\square$

4.3 Proof of both rank formulas

For even w , the column count and zero nullity give $\text{rank } A_w = 6w - 7$. Every column subfamily is independent, so deleting the $w - 1$ Gaussian columns gives $\text{rank } 5w - 6$.

For odd $w = 2m + 1$, Theorem 4.1 gives nullity $A_w = 2m = w - 1$. Thus $\text{rank } A_w = (6w - 7) - (w - 1) = 5w - 6$. A null vector of $A_w^{\neg g}$ extends by zero on the Gaussian columns to a null vector of A_w with $B = 0$. Exactly the C block remains, of dimension m . Since $A_w^{\neg g}$ has $5w - 6$ columns,

$$\text{rank } A_w^{\neg g} = 5w - 6 - m = \frac{9w - 11}{2}.$$

This proves Theorem 1.1, including the second rank separately.

5 Gaussian saturation and an exact membership algorithm

Let $\mathcal{G}_w \subset \mathcal{V}_w$ be the subspace supported on Gaussian columns. Projection of the row space onto the other columns gives

$$\dim(\mathcal{R}_w \cap \mathcal{G}_w) = \text{rank } A_w - \text{rank } A_w^{\neg g}. \quad (5.1)$$

Corollary 5.1 (Gaussian saturation in the stated module). *For odd $w \geq 3$, the Gaussian-supported row space has dimension $(w - 1)/2$ and is exactly the space spanned by the same-point product shuffles $\text{Li}_p(i) \text{Li}_{w-p}(i)$, $1 \leq p \leq (w - 1)/2$. For even w , the Gaussian-supported row space has dimension $w - 1$.*

Proof. The dimensions follow from (5.1). The indicated same-point rows belong to the Gaussian-supported space: their shuffle arguments are $(i, 1)$. To check independence without importing any period claim, write their coefficient on $g_{a, w-a}$ as

$$\binom{a-1}{p-1} + \binom{a-1}{w-p-1}.$$

For odd $w = 2m + 1$, restrict to $1 \leq a, p \leq m$. The second binomial term vanishes, and the remaining Pascal block is triangular with diagonal one. There are m independent rows in a space of dimension m . $\square$

This is stronger than merely knowing that the small shuffle system has rank m : it proves that eliminating every other color from the larger system cannot add a new Gaussian-only coefficient relation at odd weight.

Let K_w be the integer matrix whose columns are the basis vectors in Theorem 4.1. Finite-dimensional duality gives the following complete decision procedure.

Corollary 5.2 (Membership and separating certificates). *For a rational row vector t ,*

$$t \in \mathcal{R}_w \iff tK_w = 0. \quad (5.2)$$

When the product is nonzero, any column v with $tv \neq 0$ is an exact certificate $A_w v = 0$, $tv \neq 0$ against membership. In even weight every coefficient vector belongs to $\mathcal{R}_w$.

Proof. The row space is the annihilator of the nullspace over $\mathbb{Q}$. The displayed columns are a basis of that nullspace, not just sampled null vectors. $\square$

5.1 Coefficient size and construction cost

The basis can be constructed in $O(w^2)$ integer arithmetic operations. Dehomogenize with $X = 1$ and write $t = Y$. In the first block,

$$B_j(1, t) = (2t - 1)^{2j+1}, \quad E_j(1, t) = (1 - t)^{N-2j-1}(1 + t)^{2j+1}.$$

Successive B_j arise by multiplication by $(2t - 1)^2$. Successive E_j arise by two exact divisions by $1 - t$, followed by multiplication by $(1 + t)^2$. Each operation uses $O(w)$ coefficient updates. The divisions have zero remainder by the displayed factorization and are implemented by prefix sums with an explicit remainder check. The C block has

$$F_j(1, t) = (1 - t)^{N-j} - (1 - t)^j,$$

and D_j is minus its reversed coefficient vector. Binomial recurrences therefore generate this block in the same bound.

All coefficients have $O(w)$ bits: the B_j coefficients are bounded in absolute value by 3^N , the E_j coefficients by 2^N , and the remaining binomial coefficients by 2^{N+1} . A membership test then uses $O(w^2)$ integer arithmetic operations. For an integer target of coefficient height at most 2^H , a conservative schoolbook bit bound is $O(w^2(H + w + \log w)^2)$, including basis construction. Rational inputs can first be cleared of denominators; the bit size of that step must be charged to the input representation. These are bounds for the membership task, not for discovering every analytic relation among periods.

6 A uniform obstruction for the even-index harmonic sums

Define

$$S_p = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^p}, \quad u_{p,1} = \operatorname{Im} D_{p,1}(i, -1), \quad g_{p,1} = \operatorname{Im} D_{p,1}(i, 1).$$

For $p \geq 2$, absolute convergence permits separating the odd outer terms. For an outer index $2n + 1$,

$$H_{2n} + \sum_{m=1}^{2n} \frac{(-1)^m}{m} = H_n.$$

Consequently

$$S_p = g_{p,1} + u_{p,1}. \quad (6.1)$$

Theorem 6.1 (All-odd-weight obstruction). *For every odd $w = 2m + 1 \geq 3$, no rational linear combination of the rows of A_w can express S_{2m} as a rational linear combination of Gaussian doubles and the associated single-value products. More precisely, in the formal quotient $\mathcal{Q}_w$, adjoining S_{2m} to the Gaussian-coordinate span increases its dimension from m to $m + 1$.*

Proof. Choose the complementary null vector with

$$B = 0, \quad C = C_0 = Y^N - X^N, \quad N = 2m - 1.$$

It annihilates every Gaussian coordinate. Its D polynomial is $D(X, Y) = -C(Y, Y - X)$, so

$$D(1, 0) = -C(0, -1) = 1.$$

Therefore this null vector pairs to one with $u_{2m,1}$, hence also with S_{2m} in (6.1). It pairs to zero with every vector in the row space and with every Gaussian-supported vector. This excludes a reduction using the stated rows. The Gaussian span in $\mathcal{Q}_w$ has dimension m by Corollary 5.1; one new vector outside that span increases the dimension by exactly one. $\square$

6.1 An integer certificate at weight five

The source S_4 target has coefficient vector

$$T = 3g_{4,1} + 3g_{3,2} + 9g_{2,3} + 7u_{4,1}. \quad (6.2)$$

Our C_0 witness has the following nonzero coordinates:

(a, b, r, s)	value	(a, b, r, s)	value
$(1, 4, 1, 1)$	1	$(1, 4, 2, 1)$	-1
$(2, 3, 1, 2)$	3	$(2, 3, 2, 1)$	3
$(3, 2, 1, 2)$	-3	$(3, 2, 2, 1)$	-3
$(4, 1, 1, 1)$	-1	$(4, 1, 1, 2)$	1

Thus $A_5 v = 0$ and $Tv = 7$. The package records the full 92×23 matrix, all four nullspace columns, the target, and this witness. In contrast to a single isolated obstruction, the proof gives the same certificate pattern in every odd weight. The previously supplied rational witness remains valid; this is a simpler complementary-block alternative, not a correction to it.

The manuscript's proposed numerical S_4 evaluation remains open in this article. The obstruction is fully compatible with its truth. A successful proof must bring in a relation outside this fixed linear row vocabulary, or use algebraic operations not already represented by these rows.

7 The inhomogeneous compiler and exact odd-weight coordinates

We now restore the right sides. This is also an independent route from the rank theorem to explicit identities, without a search for integer relations. Set

$$\ell_k(r) = \text{Li}_k(i^r), \quad \ell_1(0) = 0$$

where the last assignment is a formal regularization used only in the completed equations. It is not the value of a convergent series. Define

$$P_{r,s}(X, Y) = \sum_{p+q=w} \text{Im}(\ell_p(r)\ell_q(s))X^{p-1}Y^{q-1}, \quad (7.1)$$

$$H_N(X, Y) = \sum_{j=0}^N X^j Y^{N-j},$$

$$S_{01} = P_{0,1} - \beta(w)H_N, \quad S_{12} = P_{1,2} + \beta(w)H_N. \quad (7.2)$$

The notation P here denotes a known product polynomial, not an unknown coordinate polynomial. Define A, B, C, D, E, F as in (3.1), now with the actual imaginary double values as coefficients, and complete the missing coefficient by

$$[X^0 Y^N]A = -g_{w-1,1} - \beta(w). \quad (7.3)$$

This is simply the missing stuffle equation with $\ell_1(0) = 0$. The same completion argument as Lemma 3.1 makes the system equivalent to the original finite rows and their retained cancellation equations.

The inhomogeneous stuffle equations are

$$A + B^\top = S_{01}, \quad C + C^\top = P_{1,1}, \quad D + F^\top = S_{12}, \quad E - E^\top = P_{1,3}. \quad (7.4)$$

The four shuffle right sides, in the order (3.4)–(3.7), are $P_{0,1}, P_{1,1}, P_{1,2}, P_{1,3}$ respectively.

Introduce the known polynomials

$$e(X, Y) = P_{0,1}(Y, X - Y) - S_{01}(X, X - Y), \quad (7.5)$$

$$d(X, Y) = S_{12}(X, Y) + P_{1,2}(X, Y - X), \quad (7.6)$$

$$b(X, Y) = P_{1,1}(Y, X - Y), \quad (7.7)$$

$$h(X, Y) = P_{1,3}(Y, Y - X) - e(Y, Y - X) + e(Y - X, Y). \quad (7.8)$$

Theorem 7.1 (Closed even-weight compiler). *For even $w \geq 2$, all the imaginary double coefficients are obtained from*

$$B(X, Y) = \frac{b(X, Y) + h(X, Y)}{2}, \quad (7.9)$$

$$C(X, Y) = \frac{1}{2}(P_{1,3}(X, -Y) + P_{1,1}(Y, X) - d(X - Y, -Y) + d(X - Y, X)), \quad (7.10)$$

$$A(X, Y) = S_{01}(X, Y) - B(Y, X), \quad (7.11)$$

$$E(X, Y) = B(X - Y, X) + e(X, Y), \quad (7.12)$$

$$F(X, Y) = C(X, X - Y) - P_{1,2}(Y, X - Y), \quad (7.13)$$

$$D(X, Y) = -C(Y, Y - X) + d(X, Y). \quad (7.14)$$

The artificial coefficient in (7.3) is discarded. Every retained value is consequently an explicit rational combination of single values and products of two single values.

Proof. Eliminating A from the first affine shuffle gives $E(X, Y) = B(X - Y, X) + e(X, Y)$. The second shuffle gives

$$B(X, Y) + B(X, X - Y) = b(X, Y).$$

Substitute the formula for E in $E - E^\top = P_{1,3}$ and use the even homogeneity degree N ; this gives

$$B(X, Y) - B(X, X - Y) = h(X, Y).$$

Adding proves (7.9).

Similarly, eliminating F gives (7.14). The last shuffle states $D(u, u - v) - D(u, v) = P_{1,3}(v, u - v)$. As N is even,

$$C(u - v, -v) = C(v - u, v) = P_{1,1}(v - u, v) - C(v, v - u).$$

Substitution therefore gives

$$\begin{aligned} 2C(v, v - u) &= P_{1,3}(v, u - v) + P_{1,1}(v - u, v) \\ &\quad - d(u, u - v) + d(u, v). \end{aligned}$$

Put $(X, Y) = (v, v - u)$ to obtain (7.10). The other formulas have already been derived. Actual values satisfy the original equations, so these necessary formulas evaluate them. Alternatively, the zero homogeneous kernel proves uniqueness of the solution. No unproved independence of the single values is needed. $\square$

Corollary 7.2 (Exact coordinates in the odd weights). *For odd $w = 2m + 1$, define*

$$B^\circ = B - \frac{1}{2}b, \quad C^\circ = C - \frac{1}{2}P_{1,1}.$$

Then $B^\circ(X, Y) = -B^\circ(X, X - Y)$ and $C^\circ(X, Y) = -C^\circ(Y, X)$. They have unique expansions in the bases (4.3) and (4.6). Every other color polynomial is recovered by (7.11) and (7.12)–(7.14). Thus precisely m Gaussian and m complementary coordinates remain in this product system.

Proof. The polynomial $P_{1,1}$ is symmetric. Hence $b(X, Y)$ is invariant under $Y \mapsto X - Y$. Subtract half of the two affine symmetrization equations for B and C . The homogeneous basis theorem then applies. The recovery formulas did not require an assumption on the parity of N . $\square$

For explicit scalar formulas, the single values used by the compiler are

$$\begin{aligned} \ell_1(1) &= -\frac{1}{2} \log 2 + \frac{\pi i}{4}, & \ell_1(2) &= -\log 2, \\ \ell_n(0) &= \zeta(n), & \ell_n(2) &= (2^{1-n} - 1)\zeta(n), \quad n \geq 2, \\ \ell_n(1) &= 2^{-n}(2^{1-n} - 1)\zeta(n) + i\beta(n), & \ell_n(3) &= \overline{\ell_n(1)}. \end{aligned}$$

Even zeta values and odd beta values are then replaced by their standard Bernoulli- and Euler-number expressions in powers of π . The implementation keeps even beta values, odd zeta values, π , and $\log 2$ as formal real atoms. Testing equality of the resulting polynomial expressions does not presume that their evaluated atoms are algebraically independent.

8 Two all-weight mixed-color identities

The polynomial compiler has particularly simple endpoint coefficients. Define

$$c_p = \operatorname{Im} D_{p,1}(i, i), \quad e_p = \operatorname{Im} D_{p,1}(i, -i).$$

These are different color patterns from $g_{p,1} = \operatorname{Im} D_{p,1}(i, 1)$.

Theorem 8.1 (Mixed leading-index families). *For every integer $m \geq 1$,*

$$c_{2m-1} = (m-1)\beta(2m) - \frac{\log 2}{2}\beta(2m-1) - \sum_{k=1}^{m-1} (1-2^{1-2k})\zeta(2k)\beta(2m-2k), \quad (8.1)$$

$$e_{2m-1} = m\beta(2m) - \frac{\log 2}{2}\beta(2m-1) - \sum_{k=1}^{m-1} \zeta(2k)\beta(2m-2k). \quad (8.2)$$

In particular,

$$e_{2m-1} - c_{2m-1} = \beta(2m) - \sum_{k=1}^{m-1} 2^{1-2k}\zeta(2k)\beta(2m-2k). \quad (8.3)$$

Proof. Set $w = 2m$, $N = w - 2$. The desired coefficients are $C(1,0)$ and $E(1,0)$. By (7.10),

$$2C(1,0) = P_{1,3}(1,0) + P_{1,1}(0,1) - d(1,0) + d(1,1).$$

The first two terms sum to $-(\log 2)\beta(w-1)$. Definition (7.6) and $H_N(1,1) = N+1$ give

$$d(1,1) - d(1,0) = P_{1,2}(1,1) - P_{1,2}(1,-1) + N\beta(w).$$

In the difference of the two product polynomials, only even inner index $q = 2k$ remains. Its coefficient is $2\beta(w-2k)\ell_{2k}(2) = -2(1-2^{1-2k})\zeta(2k)\beta(w-2k)$. Dividing by two proves (8.1).

For the second identity, (7.9) and (7.12) imply

$$2E(1,0) = P_{1,1}(1,0) + P_{1,3}(1,0) + e(1,0) + e(0,1).$$

Again the first two terms give $-(\log 2)\beta(w-1)$. Definition (7.5) yields

$$e(1,0) + e(0,1) = -P_{0,1}(1,1) + P_{0,1}(1,-1) + w\beta(w).$$

Here $\ell_1(0) = 0$ and N is even. The product-polynomial difference is $-2\sum_{k=1}^{m-1} \zeta(2k)\beta(w-2k)$. This proves (8.2). Subtraction gives (8.3). $\square$

8.1 Explicit examples

With $G = \beta(2)$, the first weights are

$$\begin{aligned} c_1 &= -\frac{\pi}{8}\log 2, & e_1 &= G - \frac{\pi}{8}\log 2, \\ c_3 &= \beta(4) - \frac{\pi^2 G}{12} - \frac{\pi^3 \log 2}{64}, & e_3 &= 2\beta(4) - \frac{\pi^2 G}{6} - \frac{\pi^3 \log 2}{64}. \end{aligned}$$

At weight six the formulas become

$$\operatorname{Im} \operatorname{Li}_{5,1}(i, i) = 2\beta(6) - \frac{\pi^2 \beta(4)}{12} - \frac{7\pi^4 G}{720} - \frac{5\pi^5 \log 2}{3072}, \quad (8.4)$$

$$\operatorname{Im} \operatorname{Li}_{5,1}(i, -i) = 3\beta(6) - \frac{\pi^2 \beta(4)}{6} - \frac{\pi^4 G}{90} - \frac{5\pi^5 \log 2}{3072}. \quad (8.5)$$

A non-endpoint example from the full compiler is

$$\operatorname{Im} \operatorname{Li}_{4,2}(i, -1) = -\frac{9}{2}\beta(6) - \frac{\pi^2\beta(4)}{48} + \frac{3\pi^3\zeta(3)}{64} + \frac{465\pi\zeta(5)}{512}. \quad (8.6)$$

The package contains all 210 retained-coordinate identities in even weights 2, 4, 6, 8, 10, 12. It separately checks both leading-index families through weight 16. These formulas are proved specializations, not PSLQ guesses. Their identification as consequences of known product and parity structures is preferable to an unsupported novelty claim for each individual number.

9 Verification, provenance, and editorial corrections

9.1 Exact verification layers

The core implementation constructs the coefficient matrix directly from (2.3)–(2.4), independently of the polynomial proof. A second routine constructs the nullspace basis using coefficient recurrences, not Gaussian elimination. For each tested weight, it verifies $A_w K_w = 0$ as an exact integer matrix identity.

For rank lower bounds, a deterministic computation reduces the integer matrices modulo the prime 65521. A nonzero minor modulo that prime is a nonzero rational minor. The explicit independent integer nullspace supplies the matching rational upper bound. Thus the resulting rank certificates are exact over $\mathbb{Q}$; they are not estimates of numerical rank and not a probabilistic extrapolation from random primes. The same method handles the non-Gaussian submatrix using the C block. The central weight-five ranks are also recomputed separately by rational elimination.

The default verification receipt records:

Verification scope	Count
All-weight matrix and deleted-column rank certificates, $2 \leq w \leq 41$	40
Odd-weight integer nullspace bases and uniform S witnesses	20
Explicit even-weight coordinate formulas, $2 \leq w \leq 12$	210
Inhomogeneous product-row polynomial residuals	816
Odd-weight affine compatibility checks, $3 \leq w \leq 13$	6
Gaussian shuffle membership checks through weight 41	420
Mixed leading-index coefficient checks through weight 16	16
Independent floating-point direct-series diagnostics	54

The first seven rows are exact arithmetic checks. The last row is a separate diagnostic and is not counted as a proof or an interval enclosure. The proof of the infinite family is Sections 3–4, not the finite extent of the verification run.

Selected exact rank results are

w	columns	rank A_w	rank A_w^{-g}	nullity A_w
2	5	5	4	0
3	11	9	8	2
4	17	17	14	0
5	23	19	17	4
6	29	29	24	0
7	35	29	26	6
15	83	69	62	14
31	179	149	134	30
40	233	233	194	0
41	239	199	179	40

9.2 Independent direct-series diagnostics

The numerical script does not use the shuffle rows to evaluate its left sides. It sums (2.1) directly with 4096 outer terms at 70 decimal working digits for 54 cases with outer index at least 3 in weights 4, 6, 8. The absolute mathematical truncation bound used for comparison is

$$\sum_{n>M} \frac{H_{n-1}^{(b)}}{n^a} \leq \begin{cases} M^{1-a} \left(\frac{1 + \log M}{a-1} + \frac{1}{(a-1)^2} \right), & b = 1, \\ \frac{b}{b-1} \frac{M^{1-a}}{a-1}, & b \geq 2. \end{cases}$$

It follows by the integral test and $H_{n-1} \leq 1 + \log n$ or $H_{n-1}^{(b)} \leq b/(b-1)$. Every observed discrepancy is smaller than this bound; the largest ratio is about 0.797. Floating-point arithmetic in this script is not outward rounded, so these comparisons are diagnostics rather than fully certified intervals. Exact polynomial row residuals provide a separate algebraic check on every delivered formula.

9.3 A local correction to the source's logical wording

The source states the two rank formulas and then calls the Gaussian-only row-space dimension an equivalent statement. Rank-nullity gives (5.1), so the pair of ranks implies the dimension of the supported row space. The difference alone does not determine the two ranks individually. The correct wording before an independent rank proof is “In particular,” not “Equivalently.” Theorem 1.1 now proves both ranks and removes the need for this overstatement.

The source's 92×23 obstruction matrix and its rational witness need no correction. The previously repaired color-swapping rule also needs no new correction: this article uses it essentially. This audit is focused on the rank section and its algebraic dependencies; it is not a fresh, exhaustive audit of all 228 manuscript pages.

9.4 Integration and trust boundaries

The supplied integration fragment replaces the last rank-conjecture subsection in 05-signet-kernels.tex. It preserves the existing labels `signed:conj:rank` and `signed:eq:rank-conjecture` as compatibility aliases, but places them on a theorem and a proved formula. The fragment includes a proof and new labels under the prefix `14rank:.` Historical arrival sources and receipts should remain unchanged as provenance.

The package includes the article source and compiled PDF, the matrix and compiler routines, the exact verifier, a separate numerical script, JSON certificates, an editorial audit, provenance metadata, and an optional integration helper that refuses to operate on an unexpected Git blob. The remote repository has not been modified. The ordinary mathematical proofs have not been checked by Lean or another proof assistant, and the article has not undergone independent peer review. These qualifications do not turn the proved matrix theorem into a numerical conjecture; they specify the verification boundary.

10 Further research questions and next proof targets

The missing S_4 relation. Test each proposed additional law against the explicit complementary C_0 witness. A law annihilating this witness cannot by itself remove the obstruction. The source’s suggested Möbius transformation $z \mapsto (1 - z)/(1 + z)$, or elimination from depth three and above, is a concrete next source of candidate rows. The task is to derive an exact law with a nonzero witness pairing and the correct single-value right side, not merely a close numerical relation.

The complementary block in every odd weight. The polynomial C now supplies explicit coordinates for the part missed by Gaussian-only searches. Determine which additional relations act on these coordinates and whether they have a uniform form. Since all mixed leading harmonic sums already detect C_0 , this is a systematic target rather than a separate unstructured search at each weight.

Other cyclotomic levels. At level four the six conjugation classes split into two uncoupled reflection blocks. At levels three, six, and eight, write the analogous color-polynomial system and determine its orbit decomposition. The objective is a proven nullspace parametrization, not a fit to a dimension sequence. Modular-form and period-polynomial methods at general level offer related structures [5], but their relation vocabulary must be matched carefully to the one used here.

The real-part sector. The real sector retains even-even color coordinates and has different conjugation signs. Its boundary completion also includes different divergent components. A complete analogous analysis would show precisely where the two-reflection simplification survives and where extra blocks arise.

Integral structure and Smith normal forms. The displayed basis is integral, but this article does not claim it is a basis of the full integral kernel lattice. Determine the lattice index and the Smith normal form of independent row submatrices. These invariants could control unavoidable denominators in rational reduction certificates and explain which small primes are exceptional.

A faster constructive positive certificate. The membership test immediately returns a separating vector when the answer is negative. When the answer is positive, extend the reflection eliminations to output an explicit sparse combination of the original matrix rows for the target. The even-weight compiler is already constructive, but a uniformly sparse, row-indexed certificate format would improve independent verification and integration into a proof assistant.

Formal verification. The purely algebraic core is a feasible first formalization target: binomial coefficient extraction, completion of one missing coordinate, homogeneous polynomial substitutions, and dimensions of two eigenspaces. The analytic shuffle and regularized boundary laws can initially be explicit hypotheses, then discharged in a separate development. This separates the finite algebra from the analytic theory rather than mixing their trusted assumptions.

Arithmetic identities beyond the formal quotient. The $w - 1$ formal odd-weight coordinates need not be independent numerical periods. Study their images under known motivic or analytic constructions, stating every additional relation or period-map assumption explicitly. The rank theorem gives a reliable coordinate system for this question, not an arithmetic independence result.

11 Conclusion

The all-odd-weight matrix conjecture is settled by a complete, uniform nullspace description. Its two ranks arise from two independent degree $w - 2$ polynomial eigenspaces, and the same calculation makes the even-weight matrix injective. This resolves the source conjecture, establishes exact Gaussian saturation for the specified product system, and provides a quadratic-arithmetic membership test with explicit separating certificates.

Retaining the right sides gives a full even-weight compiler and the mixed families (8.1)–(8.2). The complementary block explains the S_4 proof obstruction in every odd weight. The next genuinely new step for S_4 is now sharply identified: find an analytically valid law outside this module that acts nontrivially on its complementary coordinates.

A Reproduction commands and data contracts

From the package root:

```
python -m pip install -r requirements.txt
python code/verify.py
python code/numerics.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The default exact run uses `--rank-max41--even-max12`. Each coordinate in the JSON data is the tuple (a, b, r, s) with the series convention (2.1). The matrix keeps repeated nonzero rows. Kernel columns are the B_j family followed by the C_j family. All matrix and kernel entries are integers; scalar formulas use explicit rational coefficients and named real atoms. The strings are data for the bundled SymPy code, not assertions of numerical independence.

`rank_certificates.json` records pivot-column information over the fixed prime and the exact upper-bound checks. `weight5_certificate.json` contains the full central matrix and witness. `S_even_obstructions.json` records the uniform witness through weight 41. `even_weight_identities.json` contains the 210 compiler outputs. `leading_families.json` independently records the 16 endpoint comparisons. `verification_report.json` identifies software versions and measured run time; that timing is environment-specific, not an asymptotic performance claim. The numerical diagnostics have their own file and expressly separate scope.

References

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